
Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE VI

Model Development and Offline Evaluation

This lecture in one slide

VI Model Development and Offline Evaluation

Model Development and Training

Model Offline Evaluation

*why this lecture matters, and what
you can do after it*

1

Model Development and Training

Evaluating ML Models

$P(\hat{y} | x)$

[1]

Six tips for model selection

$P(y|x)$

[1]

Ensembles

$P(y|x)$

Bagging

Boosting

Stacking

[1]

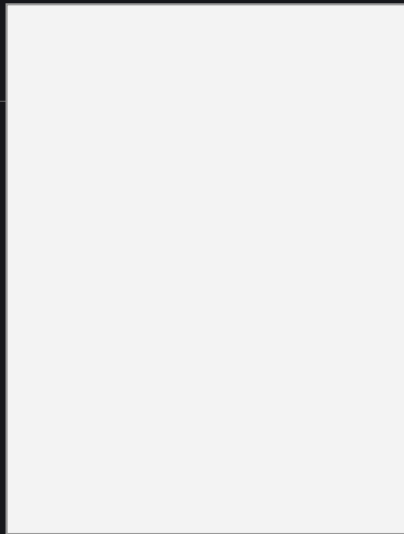
Bagging

$P(\hat{y} | x)$

Definition: Bagging

[1]

Boosting



$P(y | x)$

[1]

Stacking

$$P(\hat{y} | x)$$

[1]

Experiment Tracking and Versioning

Experiment tracking

Versioning

Experiment tracking

$P(y | x)$

[1]

Versioning

$P(y|x)$

Versioning

[1]

Distributed Training

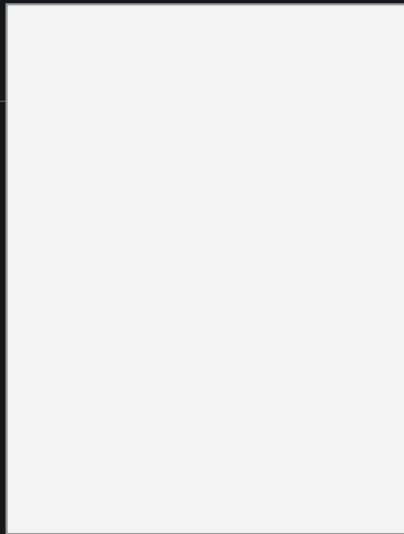
$P(y|x)$

[1]

Data parallelism

Definition: Data parallelism

Model parallelism



$P(y | x)$

[1]

AutoML

Soft AutoML

Hard AutoML

$P(y|x)$

[1]

Soft AutoML: Hyperparameter tuning

Hyperparameter tuning

$P(y|x)$

[1]

Hard AutoML: Architecture search and learned optimizer

$P(y|x)$

Architecture search

Learned optimizer

[1]

2 Model Offline Evaluation

Baselines

$P(\hat{y} | x)$

—

[1]

Evaluation Methods

$P(y|x)$

Perturbation tests

Invariance tests

**Directional expectation
tests**

Model calibration

**Confidence
measurement**

Slice-based evaluation

[1]

Perturbation tests

$P(\hat{y} | x)$

[1]

Invariance tests

Definition: Invariance tests

Directional expectation tests

$P(\hat{y} | x)$

[1]

Model calibration

$P(y | x)$

Model calibration

[1]

Confidence measurement

$$P(\hat{y} | x)$$

[1]

Slice-based evaluation

$P(\hat{y} | x)$

—

[1]

3

Summary

Summary

$P(\hat{y} | x)$

—

[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.